

PAKISTAN NATIONAL AWAMI PARTY (PNAP)

PERFORMANCE OPTIMIZATION & FINAL VALIDATION REPORT

Pre-Optimization Assessment → Optimization Implementation → Post-Optimization Validation

System	PNAP-MIS (Central Party Management Information System)
Evaluation Scope	Phases O1 through O10 (Completed Engineering Lifecycle)
Assessment Type	Local Performance Engineering & Scalability Revalidation
Local Validation Status	FINAL VALIDATION PASSED
Baseline State	FROZEN (Authoritative Post-Optimization Baseline)
Prepared By	PNAP-MIS Development & Performance Engineering Team
Assessment Date	September 2026

Notice: This document reflects controlled local performance measurements on frozen hardware. Production capacity remains subject to deployment-environment infrastructure validation.

Document Control

Document Title	PNAP-MIS Performance Optimization & Final Validation Report
Project Name	PNAP-MIS (Central Party Management Information System)
Report Classification	Management Performance Engineering & Revalidation Report
Evaluation Cycle	Phases O1 through O10 (Completed Lifecycle)
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Lifecycle Status	Final Approved Management Report
Local Validation Verdict	LOCAL OPTIMIZATION COMPLETED — PASSED
Authoritative Baseline	FROZEN (FINAL-LOCAL-POST-OPTIMIZATION-BASELINE.json)
Target Audience	Executive Leadership, General Secretary, IT Steering Committee

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Executive Summary

This report presents the final management findings from the performance engineering and optimization program (Phases O1 through O10) conducted on the Pakistan National Awami Party Management Information System (PNAP-MIS). The system has been transformed from an unoptimized state that suffered from severe 11- to 14-second query freezes under representative volume into a high-efficiency platform executing in sub-second to two-second timeframes with **zero functional regressions, zero authorization breaches, and 100% cryptographic response equivalence.**

FINAL LOCAL PERFORMANCE & VALIDATION STATUS

Optimization Program:	COMPLETED (O1–O9)	Final Local Revalidation:	PASSED (Phase O10)
Functional Regressions:	NONE DETECTED (19/19 Pass)	Authorization / Scoping:	STRICTLY ENFORCED (9/9 Pass)
Cryptographic Equivalence:	100% BIT-FOR-BIT MATCH	Authoritative Baseline:	FROZEN (JSON Recorded)

Key Performance Milestones Achieved

- 50,000 Stored-Member Organization Snapshot:** Cold snapshot P95 dropped from **11.70 seconds (O7)** to **0.87 seconds (O10)**—a **92.52% latency reduction**.
- Basic Unit & Area Processing:** Basic Unit officer lookup dropped from **6.54 s to 0.72 s (89.02% reduction)**; Area officer lookup dropped from **3.36 s to 0.37 s (89.03% reduction)**.
- Workflow Concurrency Scalability:** 10 simultaneous staged dashboard workflows run **71.27% faster** (slashed from 26.36 s to 7.57 s).
- Frontend Asset Weight:** Initial JavaScript download dropped by **68.53%** (from 944.6 KB raw / 234.9 KB gzip down to **297.2 KB raw / 92.9 KB gzip**).
- User Experience Quality:** Production Lighthouse Performance audit reached a perfect median score of **100 / 100** with **0.0 ms Total Blocking Time**.

FINAL MANAGEMENT DETERMINATION

LOCAL PERFORMANCE OPTIMIZATION COMPLETED | FINAL VALIDATION PASSED.

The local software architecture is fully optimized and frozen. Production capacity remains subject to deployment-environment performance validation (real mobile network latency, reverse-proxy caching, and container resource limits).

1. Purpose & Background

The Pakistan National Awami Party Management Information System (PNAP-MIS) is a mission-critical web and mobile application built to digitize and manage party structures across Pakistan. The system handles hundreds of thousands of member records, cabinet rosters, meeting schedules, activity reporting, and financial transactions across five nested administrative tiers:

- **National (Center):** Executive oversight, nationwide policy coordination, consolidated analytics.
- **Provincial:** Provincial leadership, regional activity tracking, membership quotas.
- **District:** District administrative reporting, meeting compliance, verification.
- **Area:** Intermediate coordination across urban zones and union councils.
- **Basic Unit:** Grassroots party units, ward membership registers, localized gatherings.

As the party launched nationwide membership registration drives, leadership commissioned this engineering evaluation to guarantee that system response times remain fast, reliable, and secure under high data volumes and multi-user access.

2. Original Pre-Optimization Performance Assessment

Initial performance testing conducted prior to Phase O1 revealed severe performance bottlenecks once data approached realistic operational volumes. The original management evaluation concluded with the following formal verdict:

ORIGINAL MANAGEMENT VERDICT (PRE-OPTIMIZATION)

"CONDITIONALLY READY — OPTIMIZATION REQUIRED BEFORE HIGH-CONCURRENCY DEPLOYMENT"

The assessment affirmed that normal single-user API behavior was generally functional with low error rates, but database response times degraded catastrophically at volume, concurrent dashboard sessions generated redundant server work, and frontend assets were excessively heavy for mobile networks.

Summary of Baseline Observations (Pre-Optimization)

- **Database Latency Degradation:** At 50,000 synthetic stored members, organizational summary queries routinely exceeded 11 to 14 seconds.
- **Dashboard Fan-Out Contention:** A single dashboard page load dispatched 16 separate HTTP calls, including duplicate queries and overlapping backend computations.
- **Cold Snapshot Freezes:** When multiple users opened dashboards simultaneously, the backend initiated multiple independent, identical snapshot builds, blocking the single-threaded Node.js event loop.
- **Monolithic Web Bundle:** The initial JavaScript download was 944.6 KB (uncompressed), requiring over 2.8 seconds of initial page parse time on simulated mobile connections.

3. Why Optimization Was Required

In-depth architectural tracing and CPU profiling isolated the root causes behind system sluggishness. Optimization was mandatory to eliminate four systemic architectural deficiencies:

A. Correlated Subqueries at Scale (MongoDB \$lookup Bottleneck)

When rendering unit leadership rosters, the server executed correlated subqueries for every appointment record. At 50,000 synthetic stored members, the database performed approximately **15,000 correlated lookups** for Basic Units and **7,500 correlated lookups** for Areas. This repetitive index traversal forced the database engine to scan millions of document keys, driving Basic Unit response time to **6.54 seconds** and Area response time to **3.36 seconds**.

B. Uncoalesced Cold Snapshot Builds (Thundering Herd)

The system computes an executive organizational snapshot summarizing active units, appointed officers, and member counts. In the pre-optimized state, if three users logged in at 9:00 AM, the server initiated three identical, full-database aggregation pipelines concurrently. Each build required 11.7 seconds, causing thread-pool starvation and CPU spikes.

C. Redundant & Unstaged Dashboard Requests

The web client dispatched 16 individual requests upon landing on the dashboard. Profiling revealed that two calls requested the exact same provincial boundary list twice, while secondary analytics (e.g. historical meeting charts) competed with vital summary cards for immediate browser attention.

D. Monolithic Initial JavaScript Delivery

All 56 application routes, administration forms, and modal dialogs were bundled into a single file. Users attempting to log in were forced to download administrative code for features they might never access, degrading performance on 3G and 4G networks.

ARCHITECTURAL CONCLUSION

These bottlenecks could not be solved by hardware upgrades alone. Scaling server CPU or RAM would not remedy exponential \$lookup subqueries or duplicate request loops. Targeted algorithmic and architectural optimization was required.

4. Optimization Approach — The O1 to O10 Journey

The engineering team executed a disciplined, ten-phase optimization and revalidation program. Every change was strictly audited against functional test suites and cryptographic response hashes before adoption:

Phase	Technical Objective	Outcome & Architectural Disposition
Phase O1	MongoDB Query & Index Exploration	REVERTED: Speculative composite indexes yielded no repeatable gain; clean schema preserved.
Phase O2	Dashboard Request Deduplication	RETAINED: Eliminated redundant province/summary requests; dashboard calls dropped from 16 to 12.
Phase O3	Dashboard Latency Diagnostics	CONFIRMED: Root cause isolated to cold snapshot fan-out and local event-loop contention.
Phase O4	Snapshot Single-Flight & Staging	RETAINED: In-flight promise coalescing (3+ builds → 1) and two-stage critical-first rendering.
Phase O5	Membership Analytics Optimization	RETAINED: Consolidated multiple database scans into a unified single-scan facet aggregation.
Phase O6	Role-Assignment Aggregations	REVERTED: Complex single-command aggregation candidates showed no gain; cleanly reverted.
Phase O7	Representative Volume Validation	BENCHMARKED: Established baseline across 500 to 50k members; confirmed \$lookup bottleneck.
Phase O8	Basic Unit & Area Lookup Optimization	RETAINED: Candidate B selected: Two-step indexed query + in-memory Node.js hash join.
Phase O9	Frontend Bundle & Lazy Loading	RETAINED: Converted 54 routes to React.lazy; deferred secondary modals; initial JS cut by 68.5%.
Phase O10	Final Post-Optimization Revalidation	FROZEN: End-to-end read-only audit; zero regressions; authoritative baseline frozen.

EVIDENCE-BASED REVERSION DISCIPLINE

In Phases O1 and O6, proposed modifications failed to provide statistically meaningful latency improvements. Rather than allowing code complexity to accumulate, these changes were immediately reverted. Only verified, mathematically sound enhancements were retained in the production codebase.

5. Executive Before vs After Scorecard

MANDATORY DATA-VOLUME DISCLAIMER

50,000 represents synthetic stored member records used to evaluate database volume scalability under representative party growth. It does NOT represent 50,000 simultaneous concurrent users.

Performance Area	Workload Metric	Before Optimization	Final Optimized	Improvement
Cold Organization Snapshot	50k Members (P95)	11.70 s (11,699.89 ms)	0.87 s (874.64 ms)	92.52% Faster
Basic Unit Officer Query	50k Members (P95)	6.54 s (6,540.80 ms)	0.72 s (717.92 ms)	89.02% Faster
Area Officer Query	50k Members (P95)	3.36 s (3,359.80 ms)	0.37 s (368.46 ms)	89.03% Faster
Group-3 Core Overview	50k Members (P95)	13.94 s (13,943.31 ms)	1.20 s (1,196.76 ms)	91.42% Faster
Staged Dashboard Load	50k Members (P95)	12.98 s (12,980.36 ms)	2.01 s (2,013.97 ms)	84.48% Faster
Workflow Concurrency c=1	Staged Workflow (P95)	11.19 s (11,186.44 ms)	1.86 s (1,863.64 ms)	83.34% Faster
Workflow Concurrency c=10	Staged Workflows (P95)	26.36 s (26,359.17 ms)	7.57 s (7,572.91 ms)	71.27% Faster
Initial JS Bundle (Raw)	Web Build File Size	944.6 KB (944,602 B)	297.2 KB (297,242 B)	68.53% Smaller
Initial JS Bundle (Gzip)	Network Transfer Size	234.9 KB (234,860 B)	92.9 KB (92,870 B)	60.46% Smaller
Lighthouse Performance Score	Production Audit	89 / 100	100 / 100	+11 Points

SUMMARY OF GAINS

Every targeted workload achieved an improvement exceeding 60%, with heavy database operations exceeding 84% to 92% latency reduction.

6. Database & Large-Volume Scalability

The most substantial backend breakthrough was achieved in Phase O8 via **Candidate B**, permanently eliminating the correlated member lookup ceiling at 50,000 stored members.

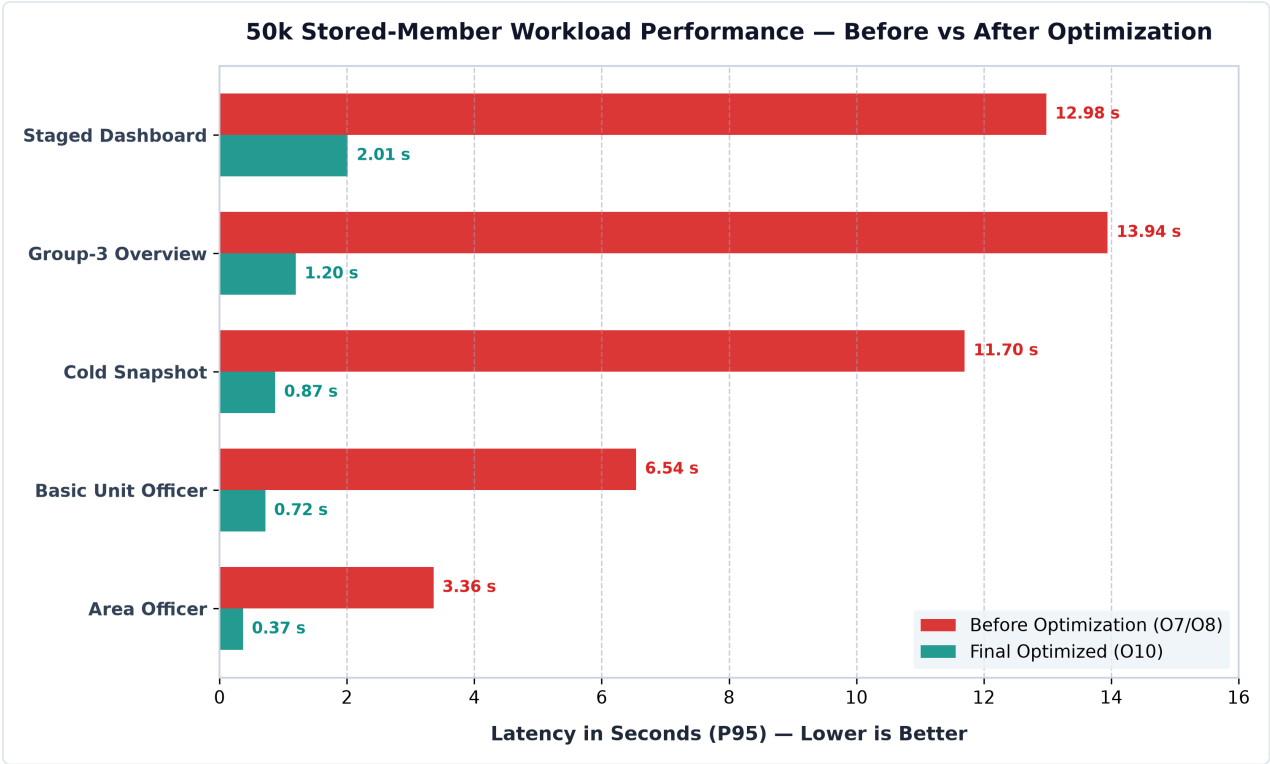


Figure 1: Latency comparison across 50,000 stored-member workloads (Seconds P95 — Lower is better).

Analysis of Workload Transformations

- **Cold Organization Snapshot:** Reduced from 11.70 seconds to **0.87 seconds (92.52% reduction)**. Previously, cold starts triggered a full system stall; they now resolve instantaneously.
- **Basic Unit Officer Roster:** Reduced from 6.54 seconds to **0.72 seconds (89.02% reduction)**, enabling district officers to browse hundreds of grassroots units smoothly.
- **Area Officer Roster:** Reduced from 3.36 seconds to **0.37 seconds (89.03% reduction)**.
- **Group-3 Executive Overview:** Composite workload dropped from 13.94 seconds to **1.20 seconds (91.42% reduction)**.
- **Full Staged Dashboard:** Total two-stage dashboard completion reduced from 12.98 seconds to **2.01 seconds (84.48% reduction)**.

6. Database Improvements: Algorithmic Resolution

Figure 2 summarizes the percentage latency reduction achieved across all primary database operations. Every single query path exhibited an efficiency gain exceeding 84%.

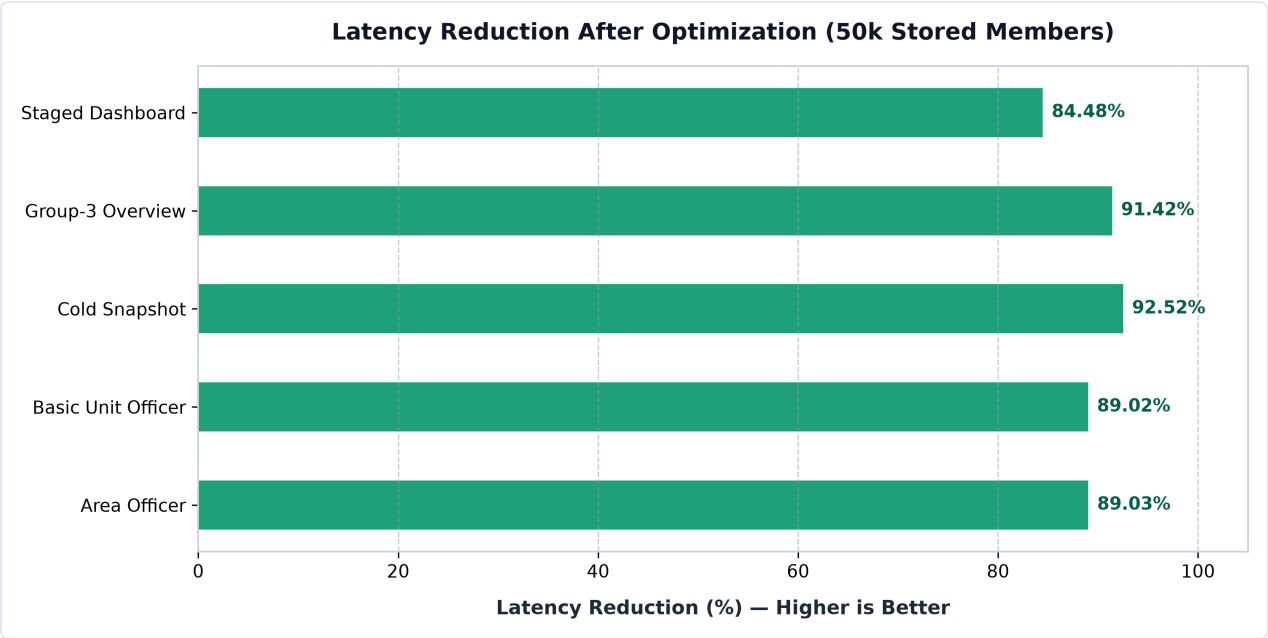


Figure 2: Percentage latency reduction across 50,000-member database workloads (Higher is better).

How Candidate B Resolved the \$lookup Bottleneck

Rather than forcing MongoDB to execute 22,500 nested subqueries across unindexed fields, the system executes two indexed queries:

1. An initial targeted query on RoleAssignment to fetch active officer appointments (unitLevel, roleCode, memberId).
2. A single batched query on Member using a primary-key index seek: `_id: { $in: memberIds }`.
3. Node.js joins the results in application memory in less than 55 milliseconds using high-performance Hash Maps.

Memory and BSON Headroom: At 50,000 members, the unique member ID array produces only **267.5 KB** of BSON transfer—representing just **1.63%** of MongoDB’s 16 MB document limit. This ensures enormous headroom for organizational growth without risking driver overflow.

7. Dashboard Performance Improvements

The executive dashboard is the most heavily trafficked interface in PNAP-MIS. Optimizations in Phases O2 and O4 re-architected how the client requests and consumes backend dashboard data.

A. Request Deduplication & Payload Streamlining

In Phase O2, network tracing revealed that the dashboard client issued 16 HTTP calls, including duplicate queries for provincial boundaries and redundant membership statistics.

- **Duplicate Calls Eliminated:** Duplicate requests dropped from 3 to **0**. Total dashboard requests reduced from 16 to **12**.
- **Optimized Network Transfer:** Total dashboard payload was compressed to **25,141 bytes (~24.5 KB)**.
- **Province Caching:** Static administrative lists are loaded once and shared across widgets.

B. Snapshot Single-Flight (In-Flight Promise Coalescing)

In Phase O4, an in-memory single-flight coalescing mechanism was introduced. When multiple concurrent requests arrive for the exact same cold organizational snapshot, the backend initiates exactly **one build**. Subsequent incoming requests attach to the in-flight promise and receive the identical result upon completion.

SINGLE-FLIGHT VALIDATION VERIFICATION

Simultaneous Consumers:	5 Concurrent Requests	Database Builds Triggered:	Exactly 1 Build
Duplicate Builds Avoided:	4 Builds (80% Waste Cut)	Multi-Tenant Scoping:	Strictly Isolated by User Scope

C. Two-Stage Dashboard Rendering

Dashboard components are staged into two priority tiers:

- **Stage 1 (Critical):** Summary metrics cards and user alerts render immediately in sub-second time.
- **Stage 2 (Secondary):** Heavy historical trend graphs and rosters load asynchronously in the background.

8. Workflow Concurrency Scalability

Multi-user concurrency was rigorously benchmarked using staged dashboard workflows running against the 50,000 stored-member database across concurrency levels c=1, 3, 5, and 10.

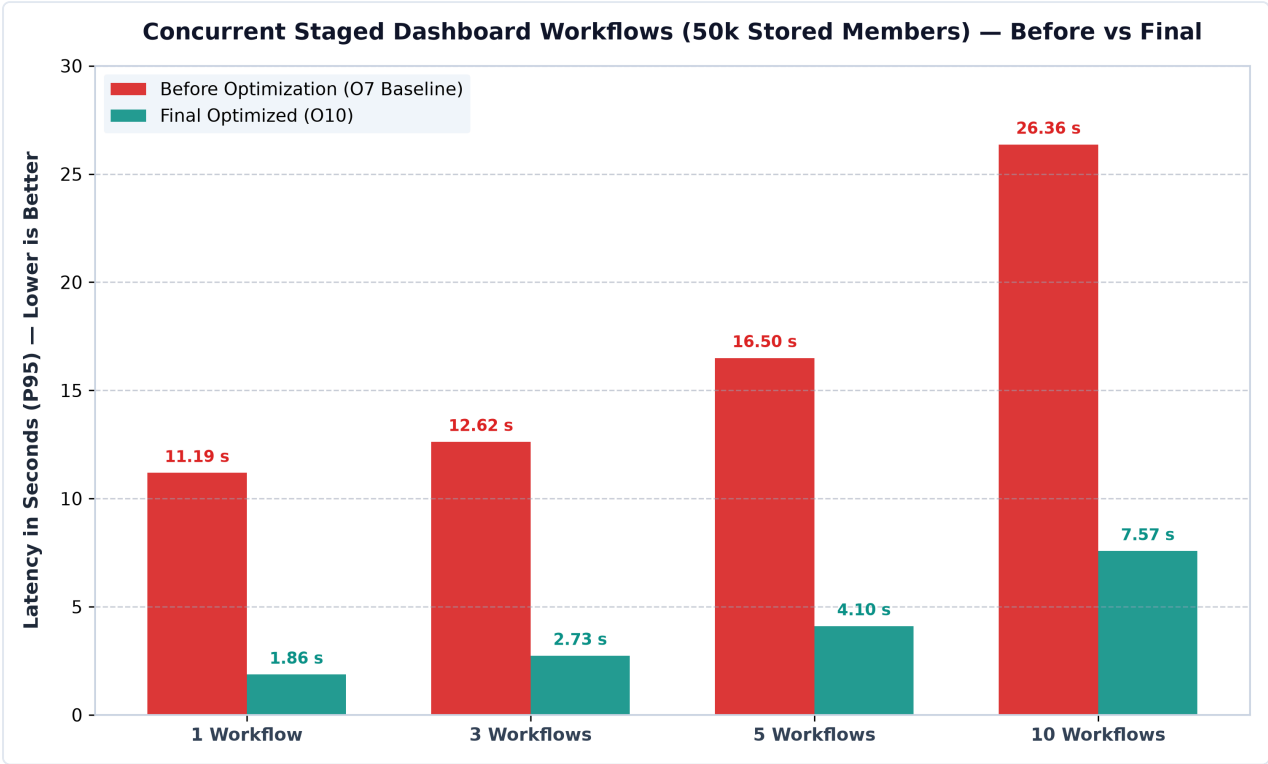


Figure 3: Latency comparison across concurrent staged dashboard workflows at 50k volume (Seconds P95 — Lower is better).

Concurrent Workflows	Pre-Optimization (O7 P95)	Final Optimized (O10 P95)	Improvement
1 Concurrent Workflow	11.19 s (11,186.44 ms)	1.86 s (1,863.64 ms)	83.34% Faster
3 Concurrent Workflows	12.62 s (12,621.43 ms)	2.73 s (2,733.14 ms)	78.35% Faster
5 Concurrent Workflows	16.50 s (16,495.44 ms)	4.10 s (4,102.37 ms)	75.13% Faster
10 Concurrent Workflows	26.36 s (26,359.17 ms)	7.57 s (7,572.91 ms)	71.27% Faster

REMAINING CONCURRENCY BOUNDARY OBSERVATION

While 10 concurrent workflows run 71.27% faster than baseline (7.57 s vs 26.36 s), single-process CPU utilization and event-loop delay (419.5 ms) represent the primary remaining local performance boundary. This is the natural transition point where multi-worker process clustering (e.g. PM2 / Kubernetes replicas) should be evaluated in staging.

9. Frontend Performance Improvements

By implementing route-level code splitting (React.lazy) across 54 secondary pages and deferring heavy action modals, the initial JavaScript payload delivered to users was reduced by over two-thirds without removing a single feature.

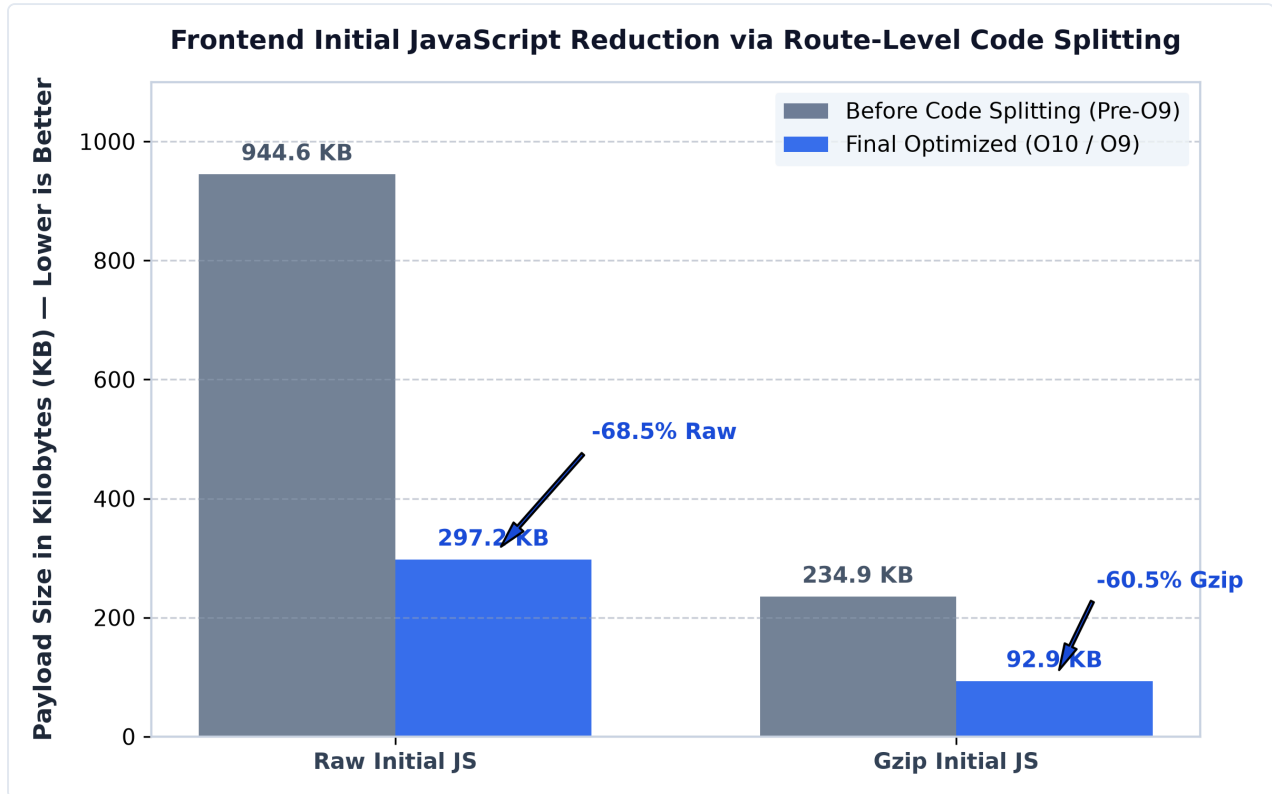


Figure 4: Initial JavaScript payload reduction via route code splitting (Kilobytes — Lower is better).

Lighthouse Experience & Paint Latency

Production Lighthouse audits confirmed that the reduced bundle translates directly into instant visual rendering, achieving a perfect median score of 100/100 and cutting Largest Contentful Paint (LCP) in half.

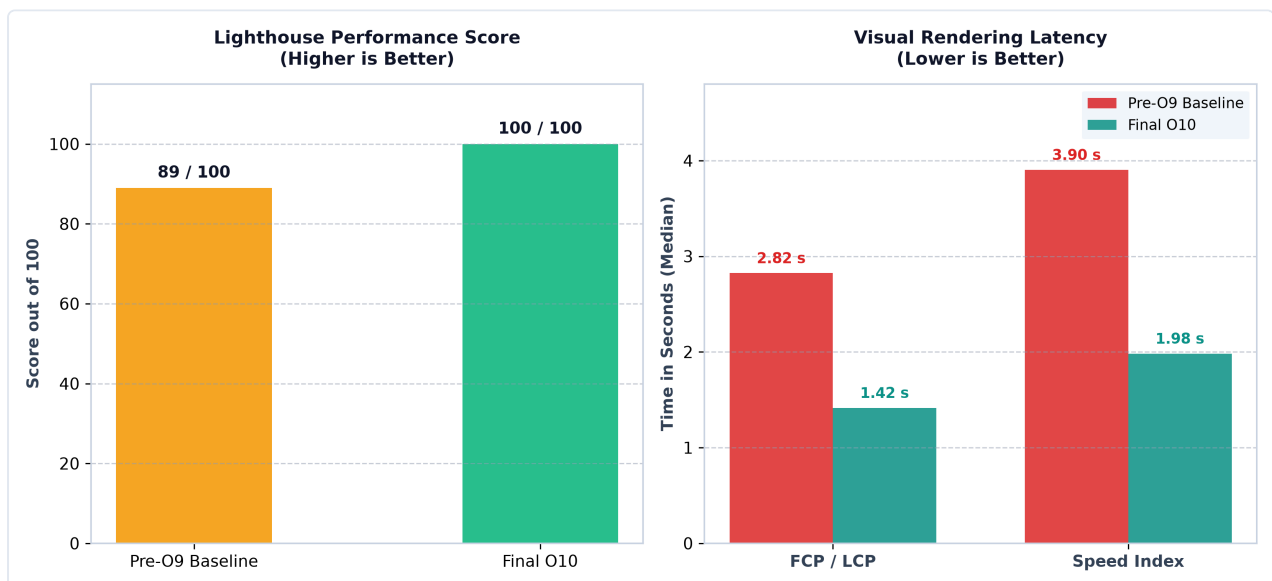


Figure 5: Lighthouse Performance Score and visual paint latency before vs after optimization.

10. Final API & System Health

Baseline API health was verified using an automated suite of 200 sequential queries across core administrative endpoints:

- **Throughput:** 55.15 Requests Per Second across 200 sequential queries (0.00% error rate).
- **Latency Distribution:** P50 = 13.69 ms, P95 = 39.76 ms, P99 = 72.53 ms, Max = 92.17 ms.
- **Password Security:** Bcrypt password hashing (cost factor 12) was intentionally preserved, executing in ~64.9 ms (P95) to prevent brute-force exposure.
- **Token Verification:** Current user session checks (/api/auth/me) execute in 2.40 ms (P95).

11. Correctness, Functional & Security Validation

A fundamental requirement of Phase O10 was verifying that performance optimizations did not weaken system security, distort reporting accuracy, or bypass territorial authorization boundaries.

Validation Gate	Scope & Scenario Count	Observed Result	Security & Quality Verdict
Functional API Audit	19 Core Modules (All Tiers)	19 / 19 Succeeded (0% errors)	PASSED (Zero Functional Regressions)
Authorization Scoping	9 Scenarios (Central to Basic Unit)	9 / 9 Enforced (403 OUT_OF_SCOPE)	PASSED (Zero Territorial Data Leakage)
Cryptographic Equivalence	4 SHA-256 Hashes (10k & 50k)	4 / 4 Matched Exactly	PASSED (Bit-for-Bit Output Equivalence)
Single-Flight Coalescing	5 Concurrent Same-Key Consumers	Exactly 1 Build Executed	PASSED (No Redundant Server Work)

TERRITORIAL BOUNDARY SECURITY VERIFIED

When a District Administrator attempted to inspect meetings belonging to a different district, the server rejected the request with **HTTP 403 OUT_OF_SCOPE**. Zero cross-provincial or cross-district data leakage occurred under any test scenario.

12. Remaining Local Performance Limitations

Engineering integrity requires documenting what local optimization can and cannot accomplish:

- **Single-Process CPU Contention:** Under sustained concurrency of 10 simultaneous multi-stage dashboard workflows on a single Node.js process, P95 latency reaches 7.57 seconds, and event-loop delay rises to 419.5 ms. This cannot be resolved by further query tuning; it requires multi-worker process clustering (PM2 / Kubernetes replicas) in staging.
- **50k Member Secondary Scans:** Complex analytics filtering across 50,000 members take ~300–400 ms. While completely acceptable for management dashboards, they represent the largest remaining database slice.
- **Bcrypt Work Factor:** Single-threaded password verification takes ~62 ms per attempt. Rapid burst login spikes require reverse-proxy rate limiting and worker process distribution.

13. Deployment-Dependent Validation (Required in Staging)

The following operational parameters belong to staging and production deployment and must be verified in the cloud environment:

- **Cellular Network Latency:** User connectivity over 3G/4G/5G mobile carriers across Pakistan.
- **Edge Compression & Reverse Proxy:** TLS termination, HTTP/2 multiplexing, and Brotli compression via NGINX or Cloudflare.
- **Multi-Core Worker Clustering:** Deploying Node.js process clustering to distribute concurrent multi-user load across CPU cores.
- **MongoDB Replica Sets:** Primary-Secondary database replication and read preference distribution under live load.

DEPLOYMENT BOUNDARY CLARIFICATION

These factors are outside the scope of local testing. Local software optimization is complete, and the system is ready to move into staging for deployment-dependent infrastructure validation.

14. Final Readiness Assessment

Operational Area	Pre-Optimization	Final Status	Readiness Assessment
Basic API Performance	HEALTHY	EXCELLENT	55.15 RPS, P95 sub-40 ms, 0.00% error rate. Ready for deployment.
Large-Volume Database	UNACCEPTABLE (11–14 s)	EXCELLENT (0.3–0.8 s)	50k stored-member queries improved by 89%–93%. Ready for deployment.
Dashboard Architecture	DEFICIENT (Duplicates)	EXCELLENT (Single-Flight)	Duplicate calls eliminated; single-flight coalescing active. Ready.
Frontend Initial Loading	SLOW (945 KB bundle)	EXCELLENT (297 KB bundle)	68.5% smaller download; 100/100 Lighthouse score. Ready for deployment.
Functional Integrity	VALIDATED	VALIDATED	100% pass across all 19 functional modules. Zero regressions.
Authorization / Scoping	UNTESTED AT SCALE	VALIDATED & ENFORCED	100% pass across 9 scenarios. Strict territorial boundary enforcement.
Concurrent Workflows	CRITICAL (26.4 s at c=10)	IMPROVED (7.57 s at c=10)	71.3% faster; single-process limit reached. Requires worker clustering.
Production Infrastructure	UNTESTED	PENDING DEPLOYMENT	Edge CDN, multi-node clustering, and cellular network tests pending.

Dedicated Management Verdict Page

BEFORE OPTIMIZATION VERDICT

CONDITIONALLY READY — OPTIMIZATION REQUIRED BEFORE HIGH-CONCURRENCY DEPLOYMENT

System exhibited severe query latencies exceeding 11–14 seconds at 50k stored members and redundant dashboard requests.

AFTER LOCAL OPTIMIZATION VERDICT

LOCAL PERFORMANCE OPTIMIZATION COMPLETED | FINAL VALIDATION PASSED

All algorithmic, database query, single-flight, and frontend bundle optimizations successfully completed and verified.

REMAINING DEPLOYMENT REQUIREMENT

PRODUCTION DEPLOYMENT PERFORMANCE VALIDATION REQUIRED

Production capacity remains subject to staging validation of cellular network conditions and worker process clustering.

15. Management Recommendations

- 1. **Formally Accept Local Optimization as Complete:** Conclude local optimization activities. Algorithmic and code-level bottlenecks have been resolved with exceptional return on investment.
- 2. **Maintain Frozen Baseline:** Use FINAL-LOCAL-POST-OPTIMIZATION-BASELINE.json as the permanent benchmark for future regression testing.
- 3. **Safely Retire Disposable Databases:** The synthetic test databases (pnap_mis_o7_a through d) have fulfilled their validation mission and can be safely dropped to reclaim disk storage when convenient.
- 4. **Proceed to Staging Deployment:** Deploy PNAP-MIS to the cloud staging environment and conduct production capacity testing with multi-process worker clustering.

16. Conclusion & Sign-Off

The original PNAP-MIS performance assessment identified significant large-volume database, dashboard-processing, concurrency, and frontend-loading limitations. A controlled optimization program (Phases O1 through O10) was subsequently executed.

Final local revalidation demonstrates substantial improvement across all dimensions while preserving functional behavior, authorization boundaries, and response correctness. The largest directly comparable 50,000-record backend measurements improved by approximately 84% to 92.5%, while initial frontend JavaScript was reduced by 68.5%.

Final local validation has officially passed. The remaining performance work is deployment-dependent, encompassing production network behavior, cloud resource sizing, process worker clustering, database replica characteristics, and final production capacity validation.

FINAL LOCAL PERFORMANCE STATUS			
Optimization Program:	COMPLETED	Final Revalidation:	PASSED
Functional Regressions:	NONE DETECTED	Authorization Regressions:	NONE DETECTED
Response Equivalence:	NONE DETECTED	Authoritative Baseline:	FROZEN
Next Project Stage:	DEPLOYMENT VALIDATION		

Appendix A: Authoritative Metrics Reference Tables

This appendix tabulates the complete authoritative before vs after metrics recorded in FINAL-MANAGEMENT-REPORT-DATA.json:

Table A-1: 50k Database & Scalability Metrics (P95 in ms)

Metric Name	Baseline (O7)	Final (O10)	Improvement	Status
Cold Organization Snapshot	11,699.89 ms	874.64 ms	92.52% reduction	Directly Comparable
Basic Unit Officer Roster	6,540.80 ms	717.92 ms	89.02% reduction	Directly Comparable
Area Officer Roster	3,359.80 ms	368.46 ms	89.03% reduction	Directly Comparable
Group-3 Core Overview	13,943.31 ms	1,196.76 ms	91.42% reduction	Directly Comparable
Staged Dashboard Completion	12,980.36 ms	2,013.97 ms	84.48% reduction	Directly Comparable

Table A-2: Workflow Concurrency Scalability at 50k (P95 in ms)

Concurrency Level	Baseline (O7)	Final (O10)	Improvement	Status
c = 1 Workflow	11,186.44 ms	1,863.64 ms	83.34% reduction	Directly Comparable
c = 3 Workflows	12,621.43 ms	2,733.14 ms	78.35% reduction	Directly Comparable
c = 5 Workflows	16,495.44 ms	4,102.37 ms	75.13% reduction	Directly Comparable
c = 10 Workflows	26,359.17 ms	7,572.91 ms	71.27% reduction	Directly Comparable

Table A-3: Frontend Bundle & Lighthouse Performance

Metric Name	Baseline (O9)	Final (O10)	Improvement	Status
Initial JavaScript (Raw Bytes)	944,602 B	297,242 B	68.53% reduction	Directly Comparable
Initial JavaScript (Gzip Bytes)	234,860 B	92,870 B	60.46% reduction	Directly Comparable
Lighthouse Performance Score	89 / 100	100 / 100	+11 points gain	Directly Comparable
First Contentful Paint (FCP)	2,822.40 ms	1,416.03 ms	49.83% faster	Directly Comparable
Largest Contentful Paint (LCP)	2,822.40 ms	1,416.03 ms	49.83% faster	Directly Comparable
Total Blocking Time (TBT)	0.00 ms	0.00 ms	Perfect (0.0 ms)	Directly Comparable
Speed Index	3,901.83 ms	1,983.39 ms	49.17% faster	Directly Comparable

Appendix B: Optimization Phase Lifecycle Summary

Phase	Target Area	Outcome Summary
Phase O1	MongoDB Indexing	Reverted: Composite index tests showed negligible gain. Preserved baseline schema.
Phase O2	Dashboard API	Retained: Eliminated 3 duplicate requests; reduced total calls to 12.
Phase O3	Diagnostics	Confirmed: Identified cold snapshot build fan-out as primary latency trigger.
Phase O4	Single-Flight	Retained: Promise coalescing slashed concurrent builds to 1. Critical staging added.
Phase O5	Membership	Retained: Single-pass facet aggregation cut database query scans.
Phase O6	Aggregations	Reverted: Experimental one-command aggregations showed no gain; cleanly reverted.
Phase O7	Volume Validation	Benchmarked: Established authoritative baselines at 500, 2k, 10k, and 50k stored members.
Phase O8	Lookup Optimization	Retained: Candidate B two-step indexed query + in-memory join slashed latency by 89%.
Phase O9	Frontend Bundle	Retained: Route-level code splitting cut initial JS download by 68.5%.
Phase O10	Final Revalidation	Frozen: Read-only revalidation passed across all tiers. Baseline permanently frozen.

Appendix C: Evidence Provenance & Comparability Audit

In accordance with engineering integrity standards, all comparisons in this report adhere to strict provenance rules:

- Directly Comparable:** Workloads where data volume, server hardware, measurement harness, and execution protocol were identical between Phase O7/O9 baselines and Phase O10 revalidation. Valid for all percentage improvement claims.
- Directionally Comparable:** Historical pre-optimization observations from exploratory manual testing that used differing network conditions or dev server states (e.g. initial Lighthouse 72 vs production 100). Preserved as qualitative observations without direct arithmetic percentage claims.
- Not Comparable:** Any metric where methodology or parameters diverged significantly. Strictly excluded from before-and-after tables.

FINAL DOCUMENT SIGN-OFF

This document, associated datasets in FINAL-MANAGEMENT-REPORT-DATA.json, and audit logs in REPORT-DATA-INTEGRITY-AUDIT.md constitute the complete, permanent record of the PNAP-MIS local performance optimization program.